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AUTOMATED TEST SYSTEM PROGRAMMABLE ATTENUATOR AND METHOD FOR PARALLEL ANALOG TIMING TESTS

Abstract

An attenuator circuit for testing electronic devices includes a first circuit having a first input, a first output, a first resistor coupled between the first input and the first output, and a second resistor coupled between the first output and a reference node, a second circuit having a second input, a second output, and a first amplifier circuit, the second input coupled to the first output, and the first amplifier circuit coupled between the second input and the second output, and a third circuit having a third input, a third output, a second amplifier circuit, and a gain control circuit, the third input coupled to the second output, the second amplifier circuit coupled between the third input and the third output, and the gain control circuit configured to adjust a gain of the second amplifier circuit.

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Background/Summary

BACKGROUND

[0001] Manufacturing electronic devices often involves testing operating parameters of fabricated devices to determine performance for individual device acceptance according to predefined specifications and/or to assess production process and/or material variations or other characteristics of a manufacturing operation. Automatic or automated test equipment (ATE) is used in high volume electronic device manufacturing for testing electrical performance of manufactured devices and integrated components thereof, referred to as electronic devices under test or DUTs. Important electrical performance characteristics include timing, such as switching speeds of transistors. High efficiency ATE testers utilize timing measurement units, referred to as time stampers (TSs), to measure the time between events or other temporally relevant information during operation of a tested electronic device. Production cost can be reduced by good parallel test efficiency (PTE) where the ATE system concurrently tests multiple DUTs. However, many ATE testers can accommodate only a limited number of TS for concurrent testing. For example, an ATE may be able to simultaneously test with a certain number of low power general purpose analog TSs but is limited to a smaller number of high speed digital TSs. In addition, certain DUTs are designed to operate at relatively high voltages that cannot be accommodated by the low power general purpose analog TSs, and cannot be economically tested with high PTE.

SUMMARY

[0002] In one aspect, an attenuator circuit includes first, second, and third circuits, where the first circuit has a first input, a first output, a first resistor coupled between the first input and the first output, and a second resistor coupled between the first output and a reference node, the second circuit has a second input, a second output, and a first amplifier circuit, where the second input is coupled to the first output, and the first amplifier circuit is coupled between the second input and the second output. The third circuit has a third input, a third output, a second amplifier circuit, and a gain control circuit, where the third input is coupled to the second output, the second amplifier circuit is coupled between the third input and the third output, and the gain control circuit is configured to adjust a gain of the second amplifier circuit.

[0003] In another aspect, a test system includes a handler interface board (HIB), having first and second sockets, and first and second attenuator circuits, where the first socket is configured to allow installation of a first device under test (DUT), and the second socket is configured to allow installation of a second DUT. The test system includes a first time stamper (TS) connected to the first socket, and a second TS connected to the second socket. The individual first and second attenuator circuits include first, second, and third circuits, where the first circuit is connected to a respective one of the first and second sockets and has a first input, a first output, a first resistor coupled between the first input and the first output, and a second resistor coupled between the first output and a reference node. The second circuit has a second input, a second output, and a first amplifier circuit, the second input coupled to the first output, and the first amplifier circuit coupled between the second input and the second output. The third circuit is connected to a respective one of the first and second TSs and has a third input, a third output, a second amplifier circuit, and a gain control circuit, the third input is coupled to the second output, the second amplifier circuit is coupled between the third input and the third output, and the gain control circuit is configured to adjust a gain of the second amplifier circuit.

[0004] In a further aspect, a method of fabricating an electronic device includes installing first and second electronic devices into respective first and second sockets of a handler interface board (HIB), adjusting a channel attenuator gain of respective first and second attenuator circuits of the HIB, and testing an electrical parameter of the first and second electronic devices using first and second time stampers that are connected to the respective first and second attenuator circuits of the HIB.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** is a schematic diagram of a test system with a programmable gain attenuator circuit.

[0006] FIG. **1**A is a schematic diagram of multiple channels of a parallel test implementation in the test system.

[0007] FIG. **2** is a flow diagram of a test method for fabricating an electronic device.

[0008] FIG. **3** is a graph of a switching signal from a tested device at an input of the attenuator circuit.

[0009] FIG. **4** is a graph of a signal from an output of the attenuator circuit.

DETAILED DESCRIPTION

[0010] In the drawings, like reference numerals refer to like elements throughout, and the various features are not necessarily drawn to scale. Also, the term “couple” or “couples” includes indirect or direct electrical or mechanical connection or combinations thereof. For example, if a first device couples to or is coupled with a second device, that connection may be through a direct electrical connection, or through an indirect electrical connection via one or more intervening devices and connections. One or more operational characteristics of various circuits, systems and/or components are hereinafter described in the context of functions which in some cases result from configuration and/or interconnection of various structures when circuitry is powered and operating. The example structures include layers or materials described as over or on another layer or material, which can be a layer or material directly on and contacting the other layer or material where other materials, such as impurities or artifacts or remnant materials from fabrication processing may be present between the layer or material and the other layer or material. Unless otherwise stated, “about,” “approximately,” or “substantially” preceding a value means +/−10 percent of the stated value. One or more structures, features, aspects, components, etc., may be referred to herein as first, second, third, etc., for ease of description in connection with a particular drawing, where such are not to be construed as limiting with respect to the claims. Various disclosed structures and methods of the present disclosure may be beneficially applied to an electronic device, manufacturing, testing, and/or operating an electronic device such as an integrated circuit. While such examples may be expected to provide various improvements, no particular result is a requirement of the present disclosure unless explicitly recited in a particular claim.

[0011] FIGS. **1** and **1**A show an example of an electronic device **100**, also referred to as an electronic DUT in a test system **101** with automatic test equipment (ATE) configured to perform timing tests of transistor components of manufactured electronic devices **100** installed into corresponding sockets using corresponding TSs **115**. The ATE system **101** has multi-stage programmable gain attenuator circuits **108** that facilitate high-volume automated production testing of manufactured electronic devices **100** with high PTE to reduce production costs. The illustrated example uses an attenuator circuit **108** and a connected TS **115** for each concurrently tested DUT **100** in a corresponding channel, with multiple channels as shown in FIG. **1**A.

[0012] The tested electronic devices **100** in the illustrated example of FIG. **1** are integrated power module devices that include a first FET transistor Q**1** and a second FET transistor Q**2** interconnected in a half bridge circuit as respective high and low side transistors. The transistors Q**1** and Q**2** each have first and second transistor terminals and a control terminal. The transistors Q**1** and Q**2** in this example are field effect transistors (FETs), where the first transistor terminal in this example is a drain D, the second transistor terminal is a source S, and the control terminal is a gate G. In other implementations, the transistor terminals and control terminal can be of a different form, such as emitter and collector transistor terminals and a base control terminal for bipolar transistors (not shown). The illustrated device **100** facilitates use in constructing power converters such as motor drives or DC-DC converters for use in a host circuit board or system. Other electronic device implementations can have more or fewer than two internal transistors or have different electronic component types that are tested for time-based performance during manufacturing.

[0013] The electronic device **100** has device terminals or leads, including a first device terminal **102** connected to the first transistor terminal D of the first transistor Q**1**, and a second device terminal **103** connected to the second transistor terminal S of the first transistor Q**1**. In the illustrated electronic device **100**, the drain of the first transistor Q**1** operates as an input voltage node labeled “VIN” with external conductivity via the first device terminal **102**. In addition, the source of the first transistor Q**1** is connected to the drain of the second transistor Q**2** to form a switching node with external access for conductivity to an output circuit by the second device terminal **103**. The electronic device **100** also has a third device terminal **104** connected to the source S of the second transistor Q**2**, which operates as a reference or ground connection labeled “GND” for operation of the electronic device **100** in a DC-DC converter circuit or system. The illustrated example also includes a fourth device terminal **105** as shown in FIG. **1** and may include further device terminals (not shown).

[0014] In operation of the test system **101**, the electronic device **100** is installed as a DUT in a socket or other suitable interconnection structure that provides electrical connection to the device terminals **102-105** for production testing of the electronic device **100**. In this example, the gate control terminal G of the first transistor Q**1** is not directly connected to any device terminal. Instead, the electronic device **100** in this example has internal control and driver circuitry including a first driver circuit DRV**1** and a control circuit **106** configured to operate the first control terminal or gate G of the first transistor Q**1**. In addition, the electronic device includes a second driver circuit DRV**2** configured to operate the second control terminal or gate G of the second transistor Q**1**.

[0015] The illustrated example also includes a communication circuit **107** connected to the control circuit **106** and having external communications connections via the fourth device terminal **105**. The communications circuit **107** in this example provides external control of the transistors Q**1** and Q**2** by connection of the control circuit **106** to the gate control terminals of the transistors Q**1** and Q**2** via the respective associated driver circuits DRV**1** and DRV**2**. The electronic device **100** may include further logic and/or control circuitry, for example, pulse width modulation switching control logic, and may include programmed and/or programmable elements (not shown) to facilitate operation as a power module to implement one or more power conversion functions when installed in a host system (not shown). The electronic device **100** (DUT) is merely one example, and any suitable type of electronic device can be tested in the system **101** to evaluate electrical performance.

[0016] As further shown in FIG. **1**A, the system **101** includes an ATE tester **110**, such as a commercially available automated tester that includes a controller **120** operatively coupled with multiple TSs **115** in each of a number of different logical sectors **123**. The system **101** also includes a handler interface board (HIB) **121** with multiple channels. Each channel of the HIB includes a DUT socket **122** and a programmable gain multistage attenuator circuit **108**. The ATE controller **120** is operatively coupled with each of the TSs **115** in order to set or adjust associated trigger or threshold levels (e.g., signal voltage thresholds) and to receive measured timestamp information or timing data, such as transistor rise times, etc. In addition, the controller **120** is operatively coupled with the individual attenuator circuits **108** as shown in FIG. **1**. In the illustrated example, the ATE controller **120** digitally programs an attenuator gain of each respective attenuator circuit **108**. The controller **120** can also operate an input switching circuit **119** to selectively apply one of multiple signals to an input stage of the attenuator circuit **108**. As further shown in FIG. **1**, the controller **120** is operatively coupled with the electronic devices **100** in order to control operation of the tested transistors Q**1** and Q**2** via the communications circuit **107** and the control circuit **106** of each installed DUT electronic device **100**.

[0017] As further shown in the example of FIG. **1**, the test system **101** includes a test capacitor C**1** connected between the device terminals **102** and **104** of each channel, as well as an output circuit **109** to provide an operational load to the half bridge circuit of the tested electronic device **100** installed in the system circuit of each channel. The output circuit **109** in one example includes an output inductor L**2**, a load capacitor C**2**, and a resistive load **116**. The ATE controller **120** in one example includes a programmed or programmable processor and an electronic memory for storing data and program instructions which, when executed by the processor, implement automated transistor rise time and frequency testing and other automated test functions suitable for testing one or more electrical parameters or operating characteristics of a tested DUT electronic device **100**.

[0018] In operation in one example, the ATE controller **120** actuates switching of one or both of the half bridge transistors Q**1** and/or Q**2** via the communication and control circuits **107** and **106** and the drivers DRV**1** and DRV**2** and measures a drain-source voltage of a selected one of the transistors Q**1** or Q**2** via the channel attenuator circuit **108** and TS **115** during switching operation. The controller **120** determines a transistor turn on rise time or turn off rise time in one example based on a time measurement from the TS **115** of the associated channel and implements similar operation in parallel for multiple installed electronic devices **100** using the associated channel attenuator circuit **108** and time stamper **115**. The DUT electronic device **100** can be characterized as passing or failing a given rise time test, for example, by the controller **120** determining a pass condition of the measured rise time is less than or equal to a specified threshold value or a failure condition if the measured rise time is greater than the threshold.

[0019] The controller **120** can perform multiple tests of each installed electronic device **100** in one example. The attenuator circuit **108** in FIG. **1** is configured by socket connections to the device terminals **102**, **103**, and **104** to measure a voltage between the first and second device terminals **102** and **103** (e.g., a drain-source voltage of the first transistor Q**1**) as well as to measure a voltage between the second and third device terminals **103** and **104** (e.g., a drain-source voltage of the second device transistor Q**2**) by selective operation of an input switching circuit **119**. In one example, the controller **120** operates the installed electronic device **100** and the TS **115** with the input switching circuit **119** in a first position to test a rise time of the first transistor Q**1**, and then operates the device **100** and the TS **115** with the input switching circuit **119** in a second position to test a rise time of the second transistor Q**2**.

[0020] In one example, the ATE **110** uses programmed digital control of an attenuator gain of the individual attenuator circuits **108** via the controller **120** in order to use a larger number of low power universal or general purpose analog TSs **115** for increased parallel test deficiency and lower manufacturing cost in producing the electronic devices **100**. In one example, the system **100** can include a Teradyne EST-800 ATE tester operatively coupled with the programmable attenuator circuits **108** of the channels of the multichannel handler interface board (HIB) **121**. The HIB **121** has two or more channels and will be described with respect to first and second sockets **122** of respective first and second channels, although any number 2 or more channels can be used in various implementations.

[0021] The first socket **122** is configured to allow installation of a first DUT electronic device **100**, and the second socket configured to allow installation of a second DUT electronic device **100**. The first TS **115** is connected to the first socket through the first attenuator circuit **108** and a second TS **115** is connected to the second socket. The ATE **110** can implement logical sectors to provide multiple TSs **115** for connection to the attenuator circuits **108** of the respective channels, for example, to measure multiple time readings of a given waveform (e.g., transistor drain-source voltage waveform rise time and fall time as well as waveform frequency or period).

[0022] As best shown in FIG. **1**, the individual attenuator circuits **108** include three stages including a first circuit **111**, a second circuit **112** and a third circuit **113**. The first circuit **111** is connected to the respective socket and has a first input. The first input of the first circuit **111** in the

illustrated example is coupled to a first pole of the input switching circuit **119** to allow selective connection to the drain D of the first transistor Q**1** or to the drain D of the second transistor Q**2**. In another example, the first input of the first circuit **111** is connected directly to one of the socket connections of the corresponding channel. In a further implementation, further switching circuitry can be provided (not shown) by which the first input of the first circuit **111** can be selectively connected to one of two or more socket connections. The first circuit **111** also has a first output, a first resistor R**1** that is coupled between the first input **119** and the first output, and a second resistor R**2** that is coupled between the first output and a reference node. In the illustrated example, the reference node is coupled to a second pole of the input switching circuit **119** that allows selective connection to the source S of the first transistor Q**1** or to the source S of the second transistor Q**2**. The first and second resistors R**1** and R**2** form a resistive voltage divider configuration with a jointing node connected to the first output. The first circuit **111** provides an attenuated first output signal at the first output with an attenuated amplitude based on the values of the first and second resistors R**1** and R**2**. In one example, the first resistor has a first resistance of approximately 100 kΩ and the second resistor R**2** has a second resistance of approximately 50 Ω.
[0023] The example second circuit **112** (e.g., second stage) provides a high input impedance load to mitigate current consumption of the attenuator circuit **108**, along with an op amp based amplifier circuit to provide fast response to input signal changes with low time constant temporal operation. In one example, the input impedance of the second circuit **112** is greater than the sum of resistances of the first and second resistors R**1** and R**2** of the first circuit **111**. The second circuit **112** has a second input, a second output, and a first amplifier circuit. The second input is coupled to the first output to receive the output of the resistive voltage divider formed by the resistors R**1** and R**2**. The first amplifier circuit is coupled between the second input and the second output. Any suitable first amplifier circuit can be used in the second circuit **112**. The example of FIG. **1** includes a third resistor R**3**, a fourth resistor R**4**, and a first op-amp **118**. The illustrated example is configured as an inverting amplifier, although not a requirement of all possible implementations and other types and forms of first amplifier circuit can be used. The first op-amp **118** has a first (e.g., inverting) input labeled "−", and a second (e.g., non-inverting) second input labeled "+", as well as an output. The third resistor R**3** is coupled between the first input "−" of the first op-amp **118** and the first output of the first circuit **111**. The fourth resistor R**4** is coupled in a feedback path between the first input "−" of the first op-amp **118** and the output of the first op-amp **118**. The second input "+" of the first op-amp **118** is coupled to the reference node, and the output of the first op-amp **118** is coupled to the third input of the third circuit **113**. In one example, the third resistor R**3** has a resistance of approximately 1 kΩ and the fourth resistor R**4** as a resistance of approximately 12.1 kΩ.
[0024] The third circuit **113** of the attenuator circuit **108** is connected to provide an attenuator output signal to a respective one of the TSs **115**. As best shown in FIG. **1**, the third circuit **113** has a third input, a third output, a second amplifier circuit, and a gain control circuit **114**. The third input is coupled to the second output to receive the output signal from the second circuit **112**. The second amplifier circuit is coupled between the third input and the third output. The gain control circuit **114** is configured to adjust a gain of the second amplifier circuit. Any suitable form of second amplifier circuit and gain control circuit can be used in the third circuit **113**. The second amplifier circuit in the example of FIG. **1** includes a fifth resistor R**5**, a sixth resistor R**6**, and a second op-amp **116**. The second op-amp **116** has inverting and non-inverting first and second inputs respectively labeled "−" and "+" and an output. The fifth resistor R**5** is coupled between the output of the first op-amp **118** and the first input "−" of the second op-amp **116**. The sixth resistor R**6** is coupled in a feedback path between the first input "−" of the second op-amp **116** and the output of the second op-amp **116**. The output of the second op-amp **116** is coupled to the third output. In one example, the fifth resistor R**5** has a resistance of approximately 332 Ω and the adjustable sixth resistor R**6** has a total end-to-end resistance of approximately 10 kΩ, and the feedback path resistance provided by the gain control circuit **114** can be digitally programmed by the controller

120. In the illustrated example, the op amps **118** and **116** have supply terminals connected to a positive reference voltage (e.g., +5 V) and a negative reference voltage (e.g., −5 V) with respect to the circuit ground (GND) at the socket terminal **104** of the tested electronic device **100**.

[0025] The gain control circuit **114** of the third circuit **113** is configured to adjust a resistance of the sixth resistor R**6**. The third circuit **113** in one example also includes a bias circuit **117** with a seventh resistor R**7** and an eighth resistor R**8**. The seventh resistor R**7** is coupled between a voltage reference node V+ (e.g., +5 V) and the second input "+" of the second op-amp **116**, and the eighth resistor R**8** is coupled between the second input "+" of the second op-amp **116** and a second reference node (e.g., labeled "GND") that is connected to the electronic device terminal **104**. In the illustrated example, the attenuator circuit **108** has an adjustable signal gain of 0.03 or more and 0.20 or less, although other suitable gain adjustment ranges can be used in other implementations. In the illustrated example, moreover, the gain control circuit **114** is digitally programmable by the controller **120** in order to adjust the gain of the second amplifier circuit., This allows the controller **120** to be programmed or otherwise configured to set an attenuation gain of the attenuator circuit **108** according to one or more criterion, including without limitation a particular type of electronic device **100** being tested, a specific component type or size of the tested DUT electronic device **100** being tested, as well as test system performance adjustments, for example, to maximize a TS input signal range to improve time performance measurement accuracy and/or precision.

[0026] As illustrated in FIG. **1**A, any number of parallel channels can be used with corresponding DUT sockets **122** and connected attenuator circuit instances **108** in order to interface a potentially high voltage tested DUT electronic device **100** to a corresponding TS **115** for highly parallel concurrent testing during manufacturing of the electronic devices **100**. As discussed above, many commercially available ATE testers can accommodate only a limited number of TSs for concurrent testing. The adjustable gain attenuator circuits **108** facilitate improved parallel testing efficiency (PTE) by allowing a larger number of low-voltage general-purpose or universal TSs to be used for concurrently testing high voltage DUT electronic devices **100**. The adjustable gain an attenuation features of the attenuator circuits **108** and the programmatic digital control thereof by the controller **120** and certain implementations allows DUT signal attenuation for time-based performance measurements by a variety of different TSs having different input voltage signal ranges. This allows selection of the TSs **115** according to parallel testing efficiency goals in a manufacturing process, thereby enhancing PTE and lowering manufacturing costs.

[0027] For example, a Teradyne ETS-800 model ATE **110** may be able to simultaneously test with up to 48 low power general purpose analog TSs per sector, but this type of TS **115** is limited to 24 V input signals. The same ATE **110** may only be able to simultaneously test with eight high-voltage digital TSs **115**, which can accommodate +/−80 V input signals. To test rise time, fall time, frequency performance, or other temporal performance parameters of high voltage switching transistors, such as those used for power conversion systems, motor drives, etc., the attenuator circuit **108** allows the use of a larger number of low voltage TSs **115** for significantly improved parallel testing efficiency, while the system can accommodate testing of both low voltage and high voltage electronic devices **100** through programmatic gain adjustment by the controller **120**. The attenuator circuit **108** in this example effectively reduces the input signals of up to 110 V down to 20 V range and therefore, allows the use of universal low voltage time stampers **115** (e.g., having an input range of 24 V) for timing measurements. In this example, the attenuator circuit **108** can support a site count of up to **96** in parallel on an ETS-800 a TE tester **110**.

[0028] The attenuator circuit **108** in certain examples has a current consumption of 0.5 mA or less, such as approximately 237.4 μA in one simulated example with a programmable gain range of 0.045 to 0.180 with 256 programming steps allowing tailoring of the operational gain to a particular DUT testing application. In addition, the attenuator circuit **108** in these or another example has a time constant of 3 ns or less. Low current consumption in the attenuator circuit **108** allows testing of DUT electronic devices **100** having a source signal under test such as a reference signal only capable of driving a very low current. In addition, using first and second amplifier circuits in the second and third circuits **112** and **113**, respectively, enhances the input impedance seen by such a signal source compared with using a simple resistive divider circuit alone. In certain implementations, moreover, the amplifier circuits can be designed with high-speed op amps **118** and **116**, such as TI OPA847 op amps, which helps the attenuator circuit **108** maintain a very low time constant at the pico-second level or below. This provides significant performance test advantages with respect to timing tests of power conversion system components designed for operation at high switching speeds, where the attenuated signal provided at the third output of the attenuator circuit **108** provides accurate electronic device performance assessment where the attenuator circuit **108** has the same time-domain specs as the input signal except the voltage swing reduction. The attenuator circuits **108** allow selection of TSs **115** to maximize parallel test efficiency for the test system **101** and easy adaptation of HIBs **121** and controller programming to test a variety of existing and new device designs.

[0029] In addition, example implementations include a digitally programmable gain control circuit **114**, for example, where one implementation includes a sixth resistor R**6** in the form of a digital potentiometer (e.g., AD5142) so that the gain can be programed digitally by the controller **120**. In such implementations, the controller **120** can adjust the output signal of the attenuator circuit **108** to the desired voltage level for the best measurement resolution on a low voltage analog time stamper **115** without any hardware changes in the system **101**. The programmable attenuator gain features of the attenuator circuit **108** provides a generic solution universally applicable to various potential input signals, which is particularly attractive to ATE hardware design. Instances of the attenuator circuit **108** can, for example, be incorporated into different HIB designs to accommodate high volume manufacturing testing of a variety of different electronic device designs to facilitate system adaptation to new designs using the same a TE **110** and TSs **115** with redesigned or updated HIBs **121** and possible reprogramming of the controller **120**.

[0030] Referring also to FIGS. **2-4**, FIG. **2** shows a method **200** for fabricating an electronic device with an included electronic device test method, which can be implemented using the test system **101** described above in connection with FIGS. **1** and **1**A. The method **200** begins at **202** in FIG. **2** with fabricating electronic devices, such as the example electronic device **100** described above in connection with FIG. **1**. At **204**, the method **200** includes installing first and second (e.g., or more) electronic devices **100** (DUTs) into respective sockets (e.g., **122**) of a handler interface board HIB (e.g., **121**).

[0031] At **206** in FIG. **2**, the method **200** includes adjusting a channel attenuator gain of respective first and second attenuator circuits **108** of the HIB **121**. In one example, the controller **120** adjusts the attenuator gain by sending a suitable digital signal to the gain control circuit **114** (e.g., digital potentiometer) to set the respective channel attenuator gain based on the type of installed DUT electronic device **100** and/or according to a particular test to be run of the installed electronic device **100**. For example, multiple timing tests can be performed at different test points of a particular installed electronic device **100** (e.g., using the input switching circuit **111** of FIG. **1**).

[0032] At **208** in FIG. **2**, the method **200** continues with concurrently testing an electrical parameter of the electronic devices **100** using the respective time stampers **115** that are connected to the attenuator circuits **108** of the HIB **121**. In one example, the controller **120** operates the installed electronic devices **100** (e.g., via the communications circuit **107** of the respective installed DUTs) and obtains timestamps or time measurements from the respective TSs **115**. In one example, as illustrated and described further below in connection with FIGS. **3** and **4**, the controller **120** operates the electronic devices **100** and the TSs **115** to measure a rise time of a selected transistor (e.g., Q**1**) of the installed electronic devices **100** at **208** in FIG. **2**.

[0033] The controller **120** determines at **210** in FIG. **2** whether additional tests are desired for the installed DUT electronic devices **100**. If so (e.g., YES at **210**), the method **200** returns to **206** in FIG. **2**. In one example, the controller **120** may switch an input switching circuit **119** to perform an additional test, and the controller **120** again adjusts the channel attenuator gain at **206** of the respective attenuator circuits **108** of the HIB **121**. In one example, the controller **120** switch is the switching state of the input switching circuit **119** to perform rise time or other temporal performance measurement of the second transistors Q**2** of the installed DUT electronic devices **100**. At **208** in this example, the method includes testing a second electrical parameter of the electronic devices **100** using the time stampers **115** (e.g., a rise time of the prospective second transistors Q**2**).

[0034] Once no further tests are desired for the installed DUT electronic devices **100** (NO at **210**), the controller **120** stores and evaluates the test results at **212** in FIG. **2**. In one example, the controller **120** compares the measured transistor rise time values obtained from the perspective TSs **115** to an acceptance threshold at **212** and determines whether a given tested DUT passed or failed the rise time test based on whether the measured rise time is less than or exceeds the threshold specification.

[0035] At **214** in FIG. **2**, the method **200** continues with the controller **120** determining whether further DUT electronic devices **100** are to be tested. If so (YES at **214**) the previously tested DUT electronic devices **100** are removed from the respective HIB sockets **122**, and the method **200** returns to **204** with installing further electronic devices **100** into the respective sockets **122**. Once the new DUT electronic devices **100** are installed, the method **200** proceeds to **206** where the controller **120** again adjusts the channel attenuator gain of the respective attenuator circuits **108** of the HIB **121**. In one example, the newly installed electronic devices **100** require a different test, and/or are of a different type that provides a different signal level to the input of the attenuator circuits **108**, and the controller **120** adjust the attenuator circuit gains to accommodate the newly installed electronic devices **100**. At **208**, the controller **120** tests electrical parameters of the newly installed electronic devices **100** using the time stampers **115**. The method **200** continues as described above for the newly installed electronic devices **100**, including any desired further tests of the installed electronic devices **100** (e.g., YES at **210**). Once no additional tests are desired for the currently installed devices (NO at **210**) and no further DUT electronic devices **100** are to be tested (NO at **214**), the method **200** is completed at **216**.

[0036] FIG. **3** shows a graph **300** with a curve **302** that shows a switching signal from a tested electronic device **100** at the first input of the attenuator circuit **108** in one example. This example illustrates measurement of device transistor rise times in the test system **101** described above. In this example, the controller **120** operates the installed DUT electronic devices **100** to cause the alternate switching of the high and low side device transistors Q**1** and Q**2** while power is applied to the electronic devices **100**. The curve **302** in FIG. **3** in this example shows the sensed drain-source voltage between the device terminals **102** and **103** during switching operation over three example switching cycles starting at an initial time T**0**, with a signal swing of approximately 50 V in one example. The associated time stamper **115** detects the time between a first time T**1** at which the sensed drain-source voltage of the tested transistor Q**1** transitions upward from 0 V to a second time T**2** at which the drain-source voltage reaches a peak value (e.g., approximately 50 V), and the time stamper **115** provides a value to the controller **120** representing the rise time **304** as the difference between T**1** and T**2**. The time stamper **115** in another example (or another TS **115** of a different sector **123** in FIG. **1**A) can also measure a period **306** between times T**1** and T**3**, for example, to determine the frequency of the switching waveform shown by curve **302** in FIG. **3**.

[0037] FIG. **4** shows a graph **400** with a curve **402** that shows an attenuated signal from the third output of the attenuator circuit **108**, which has a significantly attenuated amplitude (e.g., signal swing of less than 5 V). As shown in FIG. **4**, the signal waveform curve **402** is slightly rounded due to frequency response characteristics of the attenuator circuit **108**, but the low time constant and high input impedance of the multistage attenuator circuit **108** advantageously allows measurement of the rise time **404** (T**2**−T**1**), and the attenuated signal frequency or period **406**).

[0038] Comparing the graphs **300** and **400** in FIGS. **3** and **4**, the attenuator circuit **108** allows

accurate time-based characterization of transistor rise time, frequency, fall time, or other time-based operating parameters of a tested DUT electronic device **100** without significant waveform distortion. The attenuator circuit **108**, the test system **101**, and the method **200** and alternate implementations thereof thus provide significant advantages with respect to enhanced parallel test efficiency and reduced fabrication cost while optimizing the use of maximum parallelism with programmatic adjustability for measured signal amplitudes.
[0039] In the example of FIGS. **3** and **4**, the system allows verification of high speed switching circuits with measured input signal rise times of approximately 252 nano seconds and fall times of approximately 256 ns for a tested DUT electronic device **100** having an output signal swing of approximately 50 V (FIG. **3**), with the resulting time stamper measurement of the attenuated signal (FIG. **4**) providing a measured signal rise time above approximately 256.1 nano seconds and a fall time of approximately 260.2 nano seconds for the attenuated signal having an output swing of approximately 4.349 V.
[0040] In this example, moreover, the system can measure the attenuated signal frequency, where the DUT output signal (e.g., drain-source voltage) has a switching frequency of 1 MHz, and the system is capable of measuring the attenuated signal having a switching frequency of 996 KHz. This allows production testing to identify pass or fail status of individual DUT electronic devices **100** with high parallel test efficiency to accurately determine the quality of the manufactured electronic devices **100** in a high volume manufacturing application.
[0041] Modifications are possible in the described examples, and other implementations are possible, within the scope of the claims.

Claims

1. An attenuator circuit, comprising: a first circuit having a first input, a first output, a first resistor coupled between the first input and the first output, and a second resistor coupled between the first output and a reference node; a second circuit having a second input, a second output, and a first amplifier circuit, the second input coupled to the first output, and the first amplifier circuit coupled between the second input and the second output; and a third circuit having a third input, a third output, a second amplifier circuit, and a gain control circuit, the third input coupled to the second output, the second amplifier circuit coupled between the third input and the third output, and the gain control circuit configured to adjust a gain of the second amplifier circuit.
2. The attenuator circuit of claim 1, wherein an input impedance of the second circuit is greater than a sum of resistances of the first and second resistors.
3. The attenuator circuit of claim 1, wherein the attenuator circuit has a current consumption of 0.5 mA or less.
4. The attenuator circuit of claim 1, wherein the attenuator circuit has a time constant of 3 ns or less.
5. The attenuator circuit of claim 1, wherein the attenuator circuit has an adjustable signal gain of 0.03 or more and 0.20 or less.
6. The attenuator circuit of claim 5, wherein the gain control circuit is digitally programmable to adjust the gain of the second amplifier circuit.
7. The attenuator circuit of claim 1, wherein the gain control circuit is digitally programmable to adjust the gain of the second amplifier circuit.
8. The attenuator circuit of claim 1, wherein: the first amplifier circuit includes a third resistor, a fourth resistor, and a first op-amp, the first op-amp having first and second inputs and an output, the third resistor coupled between the first input of the first op-amp and the first output, the fourth resistor coupled between the first input of the first op-amp and the output of the first op-amp, the second input of the first op-amp coupled to the reference node, and the output of the first op-amp coupled to the third input; and the second amplifier circuit includes a fifth resistor, a sixth resistor,

and a second op-amp, the second op-amp having first and second inputs and an output, the fifth resistor coupled between the output of the first op-amp and the first input of the second op-amp, the sixth resistor coupled between the first input of the second op-amp and the output of the second op-amp, the output of the second op-amp coupled to the third output, and the gain control circuit configured to adjust a resistance of the sixth resistor.

9. The attenuator circuit of claim 8, comprising a bias circuit including a seventh resistor and an eighth resistor, the seventh resistor coupled between a voltage reference node and the second input of the second op-amp, and the eighth resistor coupled between the second input of the second op-amp and a second reference node.

10. The attenuator circuit of claim 1, wherein: the first circuit is connected to a first socket of a handler interface board (HIB) to interface a first time stamper (TS) of a test system to a first device under test (DUT) installed in the first socket; the third circuit is connected to the first TS of the test system; and the HIB further comprises: a second instance of the first circuit connected to a second socket of the HIB to interface a TS of the test system to a second DUT installed in the second socket; a second instance of the second circuit having a second input, a second output, and a first amplifier circuit, the second input coupled to the first output, and the first amplifier circuit coupled between the second input and the second output; and a second instance of the third circuit connected to the second TS of the test system and having a third input, a third output, a second amplifier circuit, and a gain control circuit, the third input coupled to the second output, the second amplifier circuit coupled between the third input and the third output, and the gain control circuit configured to adjust a gain of the second amplifier circuit.

11. A test system, comprising: a handler interface board (HIB), having first and second sockets, and first and second attenuator circuits, the first socket configured to allow installation of a first device under test (DUT), and the second socket configured to allow installation of a second DUT; a first time stamper (TS) connected to the first socket; and a second TS connected to the second socket; wherein each of the first and second attenuator circuits includes: a first circuit connected to a respective one of the first and second sockets and having a first input, a first output, a first resistor coupled between the first input and the first output, and a second resistor coupled between the first output and a reference node; a second circuit having a second input, a second output, and a first amplifier circuit, the second input coupled to the first output, and the first amplifier circuit coupled between the second input and the second output; and a third circuit connected to a respective one of the first and second TSs and having a third input, a third output, a second amplifier circuit, and a gain control circuit, the third input coupled to the second output, the second amplifier circuit coupled between the third input and the third output, and the gain control circuit configured to adjust a gain of the second amplifier circuit.

12. The test system of claim 11, wherein an input impedance of each respective second circuit is greater than a sum of resistances of the first and second resistors.

13. The test system of claim 11, wherein each respective attenuator circuit has a current consumption of 0.5 mA or less.

14. The test system of claim 11, wherein each respective attenuator circuit has a time constant of 3 ns or less.

15. The test system of claim 11, wherein each respective attenuator circuit has an adjustable signal gain of 0.03 or more and 0.20 or less.

16. The test system of claim 11, wherein each respective gain control circuit is digitally programmable to adjust the gain of the respective second amplifier circuit.

17. The test system of claim 11, wherein: each respective first amplifier circuit includes a third resistor, a fourth resistor, and a first op-amp, the first op-amp having first and second inputs and an output, the third resistor coupled between the first input of the first op-amp and the first output, the fourth resistor coupled between the first input of the first op-amp and the output of the first op-amp, the second input of the first op-amp coupled to the reference node, and the output of the first op-amp coupled to the third input; and each respective second amplifier circuit includes a fifth resistor, a sixth resistor, and a second op-amp, the second op-amp having first and second inputs and an output, the fifth resistor coupled between the output of the first op-amp and the first input of the second op-amp, the sixth resistor coupled between the first input of the second op-amp and the output of the second op-amp, the output of the second op-amp coupled to the third output, and the gain control circuit configured to adjust a resistance of the sixth resistor.

18. A method of fabricating an electronic device, the method comprising: installing first and second electronic devices into respective first and second sockets of a handler interface board (HIB); adjusting a channel attenuator gain of respective first and second attenuator circuits of the HIB; and testing an electrical parameter of the first and second electronic devices using first and second time stampers that are connected to the respective first and second attenuator circuits of the HIB.

19. The method of claim 18, further comprising: again adjusting the channel attenuator gain of the respective first and second attenuator circuits of the HIB; and testing a second electrical parameter of the first and second electronic devices using the first and second time stampers.

20. The method of claim 18, further comprising: installing third and fourth electronic devices into the respective first and second sockets of the HIB; adjusting the channel attenuator gain of the respective first and second attenuator circuits of the HIB; and testing an electrical parameter of the third and fourth electronic devices using the first and second time stampers.